

11.13 Classwork: Algebra (no calculator) — Markscheme

1. [Maximum mark: 4]

$$\text{Solve } 2(\ln x)^2 = 7 \ln x - 3.$$

$$\text{Let } u = \ln x. \text{ Then } 2u^2 - 7u + 3 = 0. \quad \mathbf{M1}$$

$$(2u - 1)(u - 3) = 0$$

$$u = \frac{1}{2} \text{ or } u = 3 \quad \mathbf{A1}$$

$$\ln x = \frac{1}{2} \Rightarrow x = e^{1/2} = \sqrt{e} \quad \mathbf{A1}$$

$$\ln x = 3 \Rightarrow x = e^3 \quad \mathbf{A1}$$

2. [Maximum mark: 5]

$$\text{Solve } \ln(x^2) = (\ln x)^2 - 3 \text{ for } x > 0.$$

$$\ln(x^2) = 2 \ln x \quad (\text{for } x > 0) \quad \mathbf{M1}$$

$$\text{Let } u = \ln x:$$

$$2u = u^2 - 3$$

$$u^2 - 2u - 3 = 0 \quad \mathbf{M1}$$

$$(u - 3)(u + 1) = 0$$

$$u = 3 \text{ or } u = -1 \quad \mathbf{A1}$$

$$\ln x = 3 \Rightarrow x = e^3 \quad \mathbf{A1}$$

$$\ln x = -1 \Rightarrow x = e^{-1} = \frac{1}{e} \quad \mathbf{A1}$$

3. [Maximum mark: 5]

$$\text{Solve } 3e^{2x} + 2 = 7e^x.$$

$$\text{Let } u = e^x, \text{ so } e^{2x} = u^2:$$

$$3u^2 + 2 = 7u$$

$$3u^2 - 7u + 2 = 0 \quad \mathbf{M1}$$

$$(3u - 1)(u - 2) = 0 \quad \mathbf{M1}$$

$$u = \frac{1}{3} \text{ or } u = 2 \quad \mathbf{A1}$$

2

$$e^x = \frac{1}{3} \Rightarrow x = \ln \frac{1}{3} = -\ln 3$$

A1

$$e^x = 2 \Rightarrow x = \ln 2$$

A1

4. [Maximum mark: 6]

Solve $\cos 2x = 3 \cos x - 2$ for $0 \leq x \leq 2\pi$.

Use $\cos 2x = 2 \cos^2 x - 1$:

$$2 \cos^2 x - 1 = 3 \cos x - 2 \quad \text{M1}$$

$$2 \cos^2 x - 3 \cos x + 1 = 0$$

Let $u = \cos x$:

$$(2u - 1)(u - 1) = 0 \quad \text{M1}$$

$$u = \frac{1}{2} \text{ or } u = 1$$

$$\cos x = \frac{1}{2}: \quad x = \frac{\pi}{3} \quad \text{A1}$$

$$x = \frac{5\pi}{3} \quad \text{A1}$$

$$\cos x = 1: \quad x = 0 \quad \text{A1}$$

$$x = 2\pi \quad \text{A1}$$

5. [Maximum mark: 5]

Solve $4^x + 16 = 5 \cdot 2^{x+1}$.

$$4^x = (2^x)^2 \text{ and } 5 \cdot 2^{x+1} = 10 \cdot 2^x.$$

Let $u = 2^x$:

$$u^2 + 16 = 10u$$

$$u^2 - 10u + 16 = 0 \quad \text{M1}$$

$$u = \frac{10 \pm \sqrt{100 - 64}}{2} = \frac{10 \pm 6}{2} \quad \text{M1}$$

$$u = 2 \text{ or } u = 8 \quad \text{A1}$$

$$2^x = 2 \Rightarrow x = 1 \quad \text{A1}$$

$$2^x = 8 \Rightarrow x = 3 \quad \text{A1}$$

Accept factoring $(u - 2)(u - 8) = 0$ in place of the quadratic formula for the second M1.

6. [Maximum mark: 4]

Solve $2x + 3 = 7\sqrt{x}$.

Let $u = \sqrt{x}$ (so $u \geq 0$ and $x = u^2$):

$$2u^2 + 3 = 7u$$

$$2u^2 - 7u + 3 = 0 \quad \text{M1}$$

$$(2u - 1)(u - 3) = 0 \quad \text{M1}$$

$$u = \frac{1}{2} \text{ or } u = 3$$

$$\sqrt{x} = \frac{1}{2} \Rightarrow x = \frac{1}{4} \quad \text{A1}$$

$$\sqrt{x} = 3 \Rightarrow x = 9 \quad \text{A1}$$